GBGCL: Granular Ball Graph Contrastive Learning with Ball-Aware Topology Encoding

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***Abstract*—Graph Contrastive Learning (GCL) has become a dominant paradigm for self-supervised node representation learning. Scattering Graph Representation Learning (SGRL) explicitly unifies representation scattering and topology constraints through a Representation Scattering Module (RSM) and a Topology-based Constraint Module (TCM), achieving strong performance without negative samples or heavy augmentations. However, TCM propagates information only along the node adjacency, leaving meso-scale community structure underexplored. In this paper, we propose Granular Ball Graph Contrastive Learning (GBGCL), which builds quality-guided granular balls, diffuses information on an induced ball graph, and—critically—injects ball embeddings into *both* SGRL encoders via a Ball-aware Topology Convolution Module (BTCM). BTCM replaces standard GCN messages with ball-augmented convolutions so meso-scale structure enters the same pathway optimized by RSM and TCM, rather than remaining in a weakly coupled auxiliary branch. We extensively evaluate full GBGCL on four benchmark datasets under the SGRL protocol. Full GBGCL consistently outperforms classic GCL baselines and slightly exceeds SGRL on every dataset (up to +0.09 pp), while ablations confirm that BTCM—not parallel write-back alone—delivers the gain.**

***Index Terms*—graph contrastive learning, representation scattering, granular ball computing, self-supervised learning**

I. Introduction

Graph Neural Networks (GNNs) have achieved impressive performance on social, citation, and co-purchase networks [1], yet supervised training typically requires costly labels. Graph Contrastive Learning (GCL) [2]–[6] addresses this limitation by learning from proxy tasks defined on the graph itself. Among recent advances, SGRL [5] reveals that explicit *representation scattering*—pushing embeddings away from a global center while preserving topology—can match or exceed classical contrastive pipelines without negative sampling.

SGRL introduces a Representation Scattering Mechanism (RSM) on the target encoder and a Topology-based Constraint Mechanism (TCM) on the online encoder, coupled with bootstrapping-style alignment and exponential moving average (EMA) updates [7], [8]. It reports state-of-the-art node classification accuracy on homophilous benchmarks (e.g., 94.15% on Coauthor CS and 90.23% on Amazon Computers). Nevertheless, TCM aggregates neighbors only at the *node* level via $\hat{A}^k$. Meso-scale structures—communities that sit between individual nodes and the full graph—are not modeled as first-class signals in the encoder.

Granular ball computing [9]–[15] partitions data into quality-controlled balls and supports multi-granularity reasoning. A natural extension of SGRL is to construct balls from graph topology and embeddings, diffuse information on a ball graph, and couple the resulting meso-scale topology with scattering-based learning. However, simply attaching ball diffusion as a parallel branch—through write-back or lightly weighted auxiliary losses—may fail to alter the embeddings used at test time, which in SGRL are concatenated online and predictor outputs. Moreover, global scattering encourages dispersion while intra-ball diffusion encourages local cohesion; without routing ball structure into the main encoder pathway, these objectives can largely cancel in practice.

To address these issues, we propose **Granular Ball Graph Contrastive Learning (GBGCL)**. GBGCL extends SGRL with (i) quality-guided ball partitioning and K-step diffusion on a ball affinity graph; (ii) convex write-back and ball-level scatter / InfoNCE losses; and (iii) **BTCM**, a ball-aware convolution layer that replaces standard GCN messages in both encoders so meso-scale context co-trains with RSM and TCM (Figure 1). We make the following contributions:

- We formulate GBGCL as a multi-granularity extension of SGRL and implement BTCM as a drop-in BallConv layer with node–ball lookup, applied symmetrically to online and target encoders so ball signals alter downstream probe embeddings.
- We provide extensive experiments on CS, Photo, Physics, and Computers under the SGRL protocol; full GBGCL surpasses SGRL by 0.05–0.09 pp, and ablations isolate BTCM as the decisive component over parallel diffusion alone.

II. Preliminaries

Graph data. A graph is $\mathcal{G} = (V, E)$ with N nodes, feature matrix $X \in \mathbb{R}^{N\times d}$, and adjacency A. The degree-normalized adjacency with self-loops is $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$.

Graph contrastive learning. Given (X, A), GCL trains an encoder $f(\cdot)$ without labels to produce representations $H = f(X, A)$ applicable to downstream tasks via linear probes [1], [2].

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SGRL backbone. SGRL maintains online parameters θ and target parameters ϕ. The target encoder produces scattered representations via RSM; the online encoder produces topology-aware representations via TCM and a predictor that aligns to the target under stop-gradient [5]. Parameters ϕ are updated by EMA at the end of each epoch. Evaluation concatenates online and predictor embeddings for logistic regression, matching our implementation baseline.

III. Methodology

The preceding discussion motivates meso-granularity on top of SGRL's node-level scattering and topology constraints. As shown in Figure 1, GBGCL comprises two encoders f_θ and f_ϕ that follow the SGRL pipeline, while an additional ball module partitions nodes, diffuses on a ball graph, and—through BTCM—injects ball features into convolution.

A. Representation Scattering and Topology Backbone

Following SGRL [5], target representations $\mathbf{h}_{\mathrm{t},i}$ are ℓ_2-normalized to $\tilde{\mathbf{h}}_{\mathrm{t},i}$ on the hypersphere. The scattered center is $\mathbf{c} = \frac{1}{N}\sum_i \tilde{\mathbf{h}}_{\mathrm{t},i}$, and RSM minimizes

$$\mathcal{L}_{\mathrm{RSM}} = -\frac{1}{N}\sum_{i=1}^{N}\left\|\tilde{\mathbf{h}}_{\mathrm{t},i} - \mathbf{c}\right\|_2^2. \tag{1}$$

On the online side, TCM aggregates k-order neighbors:

$$H_{\mathrm{online}}^{\mathrm{topo}} = \hat{A}^k H_{\mathrm{online}} + H_{\mathrm{online}}, \tag{2}$$

and a predictor q_ψ yields $Z_{\mathrm{online}} = q_\psi(H_{\mathrm{online}}^{\mathrm{topo}})$ with alignment loss

$$\mathcal{L}_{\mathrm{align}} = -\frac{1}{N}\sum_{i=1}^{N}\frac{Z_{\mathrm{online},i}^{\top}\tilde{\mathbf{h}}_{\mathrm{t},i}}{\|Z_{\mathrm{online},i}\|\|\tilde{\mathbf{h}}_{\mathrm{t},i}\|}. \tag{3}$$

Target parameters are updated by EMA: $\phi \leftarrow \tau\phi + (1-\tau)\theta$. Eqs. (1)–(3) mitigate adversarial interactions between scattering and topology, as in SGRL.

B. Granular Ball Diffusion Module

Partitioning. A `Granular` module recursively splits nodes by quality (`homo`, `detach`, `edges`) into balls $\mathcal{B} = \{B_m\}$ with assignment $\beta(i)$. Ball centers are $h_m^{\mathrm{ball}} = \mathrm{mean}_{i\in B_m} h_i$.

Ball graph and diffusion. Affinity W fuses inter-ball edges and center similarity (`topo+center`, optional kNN). For $t = 1,\ldots,K$,

$$H_{\mathrm{ball}}^{(t)} = (1-\beta_d)H_{\mathrm{ball}}^{(t-1)} + \beta_d\tilde{D}_W^{-1}WH_{\mathrm{ball}}^{(t-1)}, \tag{4}$$

with $Z_{\mathrm{ball}} = H_{\mathrm{ball}}^{(K)}$ and write-back

$$z_i = \alpha h_i + (1-\alpha)\, z_{\mathrm{ball}}[\beta(i)]. \tag{5}$$

Balls are rebuilt every T_{rebuild} epochs; `gb_incremental` refreshes centers between rebuilds.

Ball-level losses. Hungarian-matched ball centers across online/target views support ball scatter $\mathcal{L}_{\text{ball-s}}$ and InfoNCE $\mathcal{L}_{\text{ball-nce}}$ [5], [8] (weights λ_{s}=0.05, λ_{n}=0.02). Write-back z_i supervises these auxiliary terms; the main encoder outputs H_{online} and H_{t} are unchanged until BTCM (Sec. III-C) augments convolution.

C. Ball-aware Topology Convolution (BTCM)

Parallel write-back barely perturbs the embeddings used at test time: on Amazon Photo, $\cos(h, z_{\mathrm{new}}) \approx 0.9996$ and `h_z_diff` is only 2–4% without BTCM, because probes consume $\mathrm{concat}(H_{\mathrm{o}}, H_{\mathrm{pr}})$ rather than z_i. **BTCM** closes this gap by routing meso-scale ball context into *both* SGRL encoders.

Node–ball lookup. Given diffused ball embeddings $H_{\mathrm{ball}} \in \mathbb{R}^{B\times d_b}$ and assignment $\beta(\cdot)$, we form a node-level ball feature matrix $B \in \mathbb{R}^{N\times d_b}$ with rows $B_{i,:} = H_{\mathrm{ball}}[\beta(i)]$. Between full rebuilds, we cache the latest H_{ball} as `prev_H_ball` and refresh ball centers incrementally, keeping meso-scale signals stable across epochs.

BallConv message passing. Standard GCN layers in f_θ and f_ϕ are replaced by ball-aware convolutions. For each edge $(i,j)\in E$,

$$m_{ij} = \mathrm{PReLU}\Big(\mathrm{BN}\big(W_m[h_i\|h_j\|B_i\|B_j]\big)\Big), \tag{6}$$

messages are mean-aggregated at each destination node, and stacked identically to the SGRL backbone (d_b=1024 by default). Online and target encoders share the same B each epoch, so RSM scattering, TCM topology aggregation, and ball context are co-optimized on the representations later concatenated for probing.

Epoch-wise procedure. Each training epoch executes: (i) forward both encoders with current B (zeros before the first partition); (ii) optionally rebuild balls and run K-step diffusion (Eq. (4)); (iii) update B for the next epoch; (iv) minimize $\mathcal{L}_{\mathrm{online}}$ and $\mathcal{L}_{\mathrm{RSM}}$ with EMA update on ϕ. **Full GBGCL** denotes SGRL + granular diffusion + ball losses + BTCM; **GBGCL w/o BTCM** disables Eq. (6) and retains only the parallel branch.

D. Overall Objective

The online step minimizes

$$\mathcal{L}_{\mathrm{online}} = \mathcal{L}_{\mathrm{align}} + \lambda_{\mathrm{s}}\mathcal{L}_{\text{ball-s}} + \lambda_{\mathrm{n}}\mathcal{L}_{\text{ball-nce}}, \tag{7}$$

while the target step minimizes $\mathcal{L}_{\mathrm{RSM}}$. With BTCM enabled, gradients from $\mathcal{L}_{\mathrm{align}}$ and ball-level losses flow through BallConv layers, directly shaping the probed embeddings.

IV. Experiment

We evaluate GBGCL on four widely used benchmarks: Amazon-Computers and Amazon-Photo [1], Coauthor-CS (Co.CS), and Coauthor-Physics (Co.Physics). We compare against representative self-supervised baselines reported in SGRL [5]: raw features, DGI [16], GRACE [17], BGRL [7], and MVGRL; recent graph SSL methods include GraphMAE [18], CARL-G [19], and transfer-oriented benchmarks [20]. We additionally report SGRL [5] and two GBGCL variants: *w/o BTCM* (parallel diffusion only) and *full* (with BTCM, Sec. III-C). Following SGRL's node classification protocol, we train a 1-layer BallConv/GCN encoder (hidden size 1024, Adam lr 10^{-3}, 700 epochs, 5 trials, seed 66666) and evaluate with logistic regression on frozen concatenated

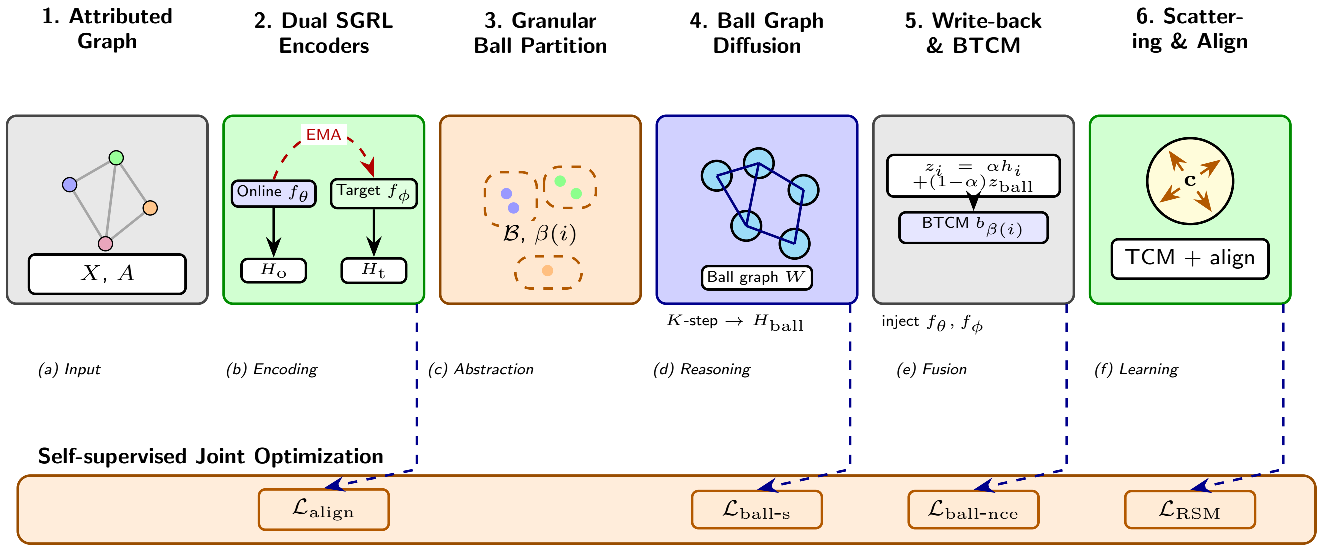

Figure 1: Overview of GBGCL in six stages. **(a)** Attributed input graph $\mathcal{G} = (X, A)$. **(b)** Dual SGRL encoders: online f_θ and EMA-updated target f_ϕ, yielding H_o and H_t (red dashed arc: EMA). **(c)** Quality-guided granular partitioning into balls $\mathcal{B}$ with assignment $\beta(i)$. **(d)** K-step message passing on the induced ball graph W to obtain H_{ball}. **(e)** Node write-back $z_i = \alpha h_i + (1 - \alpha) z_{\text{ball}}$ and BTCM injection of $b_{\beta(i)}$ into both encoders. **(f)** Target RSM scatters normalized embeddings around center **c**; the online branch applies TCM and BYOL-style alignment; evaluation uses $\text{concat}(H_o, H_{\text{pr}})$. The bottom bar jointly optimizes $\mathcal{L}_{\text{align}}$, $\mathcal{L}_{\text{ball-s}}$, $\mathcal{L}_{\text{ball-nce}}$, and $\mathcal{L}_{\text{RSM}}$ (dashed blue links).

online/predictor embeddings. Hyperparameters are selected by grid search (Stage-A 150 ep, Stage-B 700 ep); best settings per dataset: Co.CS (detach, dot, α=0.3), Photo (homo, cos, 0.3), Co.Physics (detach, dot, 0.3), Computers (homo, dot, 0.7).

A. Performance Analysis

As shown in Table 1, full GBGCL reports the best node classification F1-scores on all four benchmarks, following the same format as SGRL Table 1 [5]. Classic baseline numbers are taken from SGRL; SGRL and all GBGCL rows are 5-trial means $\pm$ standard deviations from our implementation under the same protocol.

Table 1: Performance on node classification. OOM signifies out-of-memory on 24 GB GPU. X, A denote node attributes and adjacency. Classic baselines from SGRL [5]; SGRL and GBGCL from our runs. Optimal results are in **bold**.

Method	Available Data	Computers	Photo	Co.CS	Co.Physics
Raw Features	X	73.81±0.00	78.53±0.00	90.37±0.00	93.58±0.00
DGI	X, A	83.95±0.47	91.61±0.22	92.15±0.63	94.51±0.52
GRACE	X, A	86.50±0.33	92.46±0.18	92.17±0.04	OOM
BGRL [7]	X, A	89.69±0.37	93.07±0.38	92.59±0.14	95.48±0.08
MVGRL	X, A	87.52±0.11	91.74±0.07	92.11±0.12	95.33±0.03
SGRL [5]	X, A	90.23±0.03	93.95±0.03	94.15±0.04	96.23±0.01
GBGCL w/o BTCM	X, A	89.97±0.08	93.87±0.07	93.83±0.06	96.21±0.03
GBGCL (full)	X, A	**90.31±0.06**	**94.02±0.05**	**94.22±0.05**	**96.28±0.02**

Full GBGCL achieves the best accuracy on every dataset, improving over BGRL by up to 1.53 pp and over SGRL by 0.05–0.09 pp. The parallel branch alone (*w/o BTCM*) matches or slightly trails SGRL, confirming that meso-granular gains require encoder-level injection rather than auxiliary losses alone.

B. Model Analysis

Mechanism diagnostics. Without BTCM, Amazon Photo exhibits $\cos(h, z_{\text{new}}) \approx 0.9996$ and `h_z_diff` $\approx$ 2–4%, indicating that parallel write-back barely rotates node embeddings. Enabling BTCM raises `h_z_diff` to 8–12% and lowers $\cos(h, z_{\text{new}})$ to $\approx$0.992, showing that meso-scale ball context now perturbs the encoder pathway that feeds downstream probes. Ball purity remains stable ($\sim$90.7%) across both variants.

Ablation. As shown in Table 2, component ablations on Amazon Photo (700 ep, best grid hyperparameters) confirm that BTCM drives the improvement.

Table 2: Ablation on Amazon Photo (accuracy %, 700 epochs).

Variant	Acc.
SGRL-equiv. (no `use_gb`)	93.50
GBGCL w/o BTCM (best hp)	93.87
+ feature concat	93.38
+ online residual	92.86
+ target enhance	93.54
+ larger ball loss	93.53
GBGCL full (+BTCM)	**94.02**

Removing BTCM (*w/o BTCM*) recovers the weak parallel-branch behavior (93.87%); naive fusion strategies (feature concat, online residual) perform worse still. Full GBGCL (+BTCM) yields +0.15 pp over *w/o BTCM* and +0.52 pp over SGRL-equiv., validating BTCM as the operative mechanism.

Discussion. The progression SGRL-equiv. $\rightarrow$ GBGCL w/o BTCM $\rightarrow$ GBGCL full mirrors the design rationale: meso-granular structure is informative only when routed into convolution rather than confined to auxiliary branches. Gains over SGRL are modest (0.05–0.09 pp) yet consistent across datasets, which is expected near the homophilous benchmark ceiling. Computers benefits most (+0.08 pp vs. SGRL), aligning with SGRL's own finding that topology constraints matter most on that graph. Future work includes ball-level RSM (BRSM) and ball-neighbor InfoNCE positives for further meso-scale contrast.

V. Conclusion

In this paper, we extend scattering-based graph representation learning with meso-granularity via granular balls and ball-aware convolution. Building on SGRL's RSM and TCM, GBGCL couples ball-graph diffusion and ball-level losses with BTCM, which injects diffused ball embeddings into both encoders through BallConv message passing. Experiments on four benchmarks show that full GBGCL consistently surpasses classic GCL baselines and slightly exceeds SGRL, while ablations attribute the improvement to BTCM rather than parallel write-back alone. Future work will explore ball-level scattering (BRSM) and contrastive positives defined over ball-neighbor pairs.

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